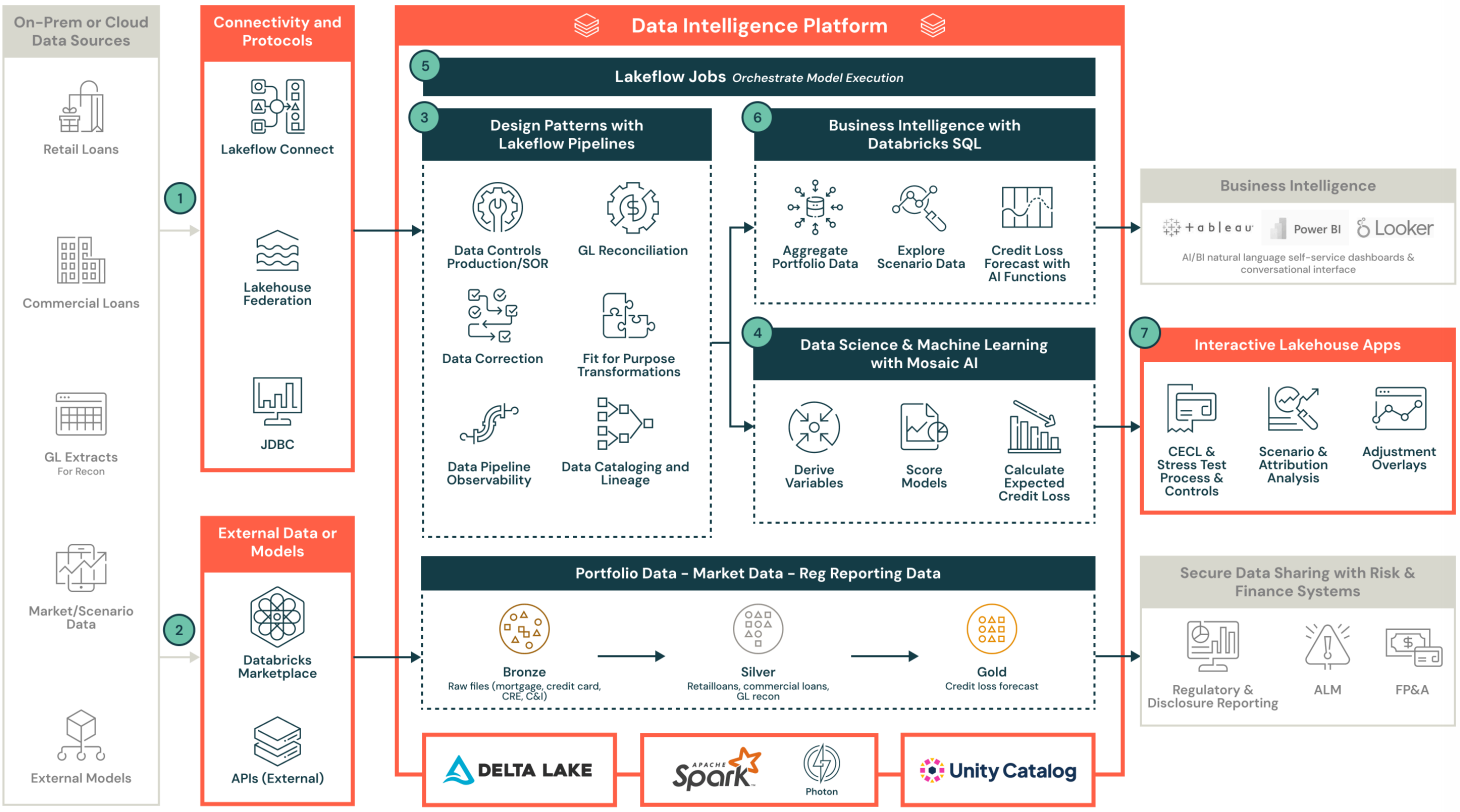
Reference Architecture for Credit Loss Forecasting

Unify loan portfolios, economic scenarios and risk models on the Databricks Data Intelligence Platform to power scalable, transparent, auditable and cost-efficient CECL and stress testing.



What you'll learn:

- An end-to-end lakehouse architecture for ingesting and unifying retail loans, commercial loans, general ledger (GL) and macroeconomic scenario data
- How Lakehouse Federation and Lakeflow Connect support secure, scalable and data integration across cloud and on-premises systems
- The use of lakehouse to standardize, reconcile and quality-check data for downstream model execution
- How to operationalize models built in Python, R or SAS using Mosaic AI and orchestrate workflows with Databricks Workflows
- A scalable compute layer using Databricks clusters to support large-scale CECL and stress testing
- A centralized data and model catalog, security model and controls with Unity Catalog to enforce data lineage, auditability and regulatory compliance
- How Lakehouse Apps enables secure collaboration, adjustments and forecast sign-off between credit risk and finance teams

Modernize Your Credit Loss Forecasting for CECL and Stress Test

1. Portfolio data sources and ingestion

- Use **Lakeflow Connect** for native, CDC-based ingestion from on-premises or cloud data systems, or leverage **Lakehouse Federation** for secure, scalable and duplication-free data access

2. **Macroeconomic scenario data**
   - Connect and source macroeconomic scenario data, such as Moody's scenarios, via APIs
   - Incorporate custom scenario expansion logic or ingest internal scenario datasets directly into the platform
3. **Data governance and management**
   - Utilize **Unity Catalog** to centralize metadata governance across portfolio data, scenario data, model outputs, overlays and disclosure reporting. **Lineage tracking** ensures data reliability and audit readiness.
   - Enable multi-asset class integration, standardizing retail and commercial loan data with row-level access controls
   - Perform data quality checks and GL reconciliation into curated Silver tables and sign off on data controls
   - Leverage **system tables** and embedded audit trails for full auditability and compliance with regulatory standards
4. **Implement model execution**
   - Implement or import models developed in Python, R or SAS. Register models in **MLflow.**
   - Define logic for variable derivation, model scoring and expected credit loss (ECL) calculations by scenario and time horizon
5. **CECL and stress test workflows**
   - Build **workflows** for scenario analysis, sensitivity analysis and attribution analysis
   - Execute workflows at scale using Databricks Workflows, providing automation, monitoring and scheduling for complex model pipelines
6. **Business intelligence**
   - Use **Databricks SQL** to review and analyze portfolio data and scenario data
   - Conduct loan-level credit loss analysis for each scenario and horizon
   - Explore results interactively and validate assumptions with full transparency and traceability
7. **Credit risk and finance collaboration**
   - Enable real-time collaboration between credit risk and finance teams via **Lakehouse Apps** (web applications)
   - Upload end user computing spreadsheets to support individual assessments
   - Apply management overlays and sign-off controls, and integrate with downstream risk and finance systems for GL posting, disclosure reporting and more

Benefits

1. **Regulatory compliance and auditability**
   Ensure compliance with CECL, CCAR, IFRS 9 and other regulatory frameworks through **automated data lineage, embedded controls** and **audit-ready workflows**
2. **Scalable performance for complex calculations**
   Run credit loss models and scenarios with ease using autoscaling **Databricks clusters** designed for compute-intensive financial workloads
3. **Cost-efficient architecture**
   Leverage a **consumption-based pricing model** with no additional software licensing fees — resulting in **lower TCO** and flexible resource usage aligned to your demand
4. **Secure, enterprise-ready platform**
   Built-in **security, identity management** and **governance** capabilities ensure sensitive risk data is protected and managed in accordance with enterprise and regulatory standards
5. **Self-service with full customization**
   Enable internal teams to **own and adapt their modeling environment** through a self-service platform, while still supporting full customization, automation and integration with enterprise systems

Recommended

INDUSTRY ARCHITECTURE
Financial Services Investment Management Reference Architecture

REFERENCE ARCHITECTURE
Intelligent Data Warehousing on Databricks

REFERENCE ARCHITECTURE
Data Ingestion Reference Architecture

REFERENCE ARCHITECTURE
Data Intelligence End-to-End Architecture with Azure Databricks

INDUSTRY ARCHITECTURE
Distribution Optimization Reference Architecture for Insurance

INDUSTRY ARCHITECTURE
Underwriting Analytics Reference Architecture for Insurance

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